

Consumption Based Responsibility for Global Warming and the Distribution of Impacts of Swiss Climate Policy

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Abstract

Different consumption patterns have been linked to different levels of responsibility for current greenhouse gas (GHG) emissions and it is well established that the affluent are responsible for higher levels of global greenhouse gas (GHG) emission than are the poor. Here I couple a life cycle assessment of consumer goods with household survey data about consumption patterns to arrive at household level responsibility for global greenhouse gas (GHG) emissions by consumption category. This allows me to provide a detailed analysis of how different consumption categories contribute to this responsibility. From this, I offer some insights into how this information can be used for the design of policies that create equitable outcomes.

I conclude that the distributional impacts of greenhouse gas (GHG) pricing with revenue recycling will remain unproblematic as climate policy continues to cover more greenhouse gas (GHG)s from more regions. If it is desired that high-income households make a bigger contribution to the emissions reduction effort than others, focusing climate policy on transport (high confidence), eating out, and clothing (both with lower confidence) may provide avenues for achieving that. This is the case, since responsibility for greenhouse gas (GHG) emissions from these consumption categories increases faster with income than it does for other goods.

Keywords: Consumption based responsibility, Greenhouse gases, Households, Distribution

1 Introduction

The distribution of consumption based responsibility for climate change of different parts of the population has recently gained increased attention. Analyses of this distribution can be used to gauge how important it is that different parts of the population partake in the decarbonization effort. The World Inequality Lab of the Paris School of Economics, e.g., observe that the rich part of the world are responsible for a share of global greenhouse gas (GHG) emissions that is disproportionate to their share of the global population. They make the argument that the additional emissions caused by lifting everybody above the poverty threshold of earning US\$ 5.5 per day, could be compensated by the top 10 percent of global emitters cutting their emissions by one third ([Chancel, Voituriez, & Bothe, 2023](#)). Similar observations can be made within countries: The more affluent consume more and therefore are responsible for a higher share of global GHG emissions than are low-income households. But apart from the general demand that high-income individuals contribute accordingly to the emissions reduction effort, little concrete policy advice has been deduced from these observations on the national level. With this paper I want to contribute to the discussion with a detailed analysis of consumption based responsibility for global GHG emissions by Swiss households.

Upon a closer look, two strands of the discussion around distributional issues can be identified. The first strand, happening more in the academic literature and finding its way into the policy discussions from there, focuses on exposure of households to costly climate policy. It finds that low-income households spend a larger fraction of their expenditure budget on energy and emissions-intensive goods than do high-income ones ([Stern, 2012](#), e.g., focuses on transport fuels). This makes them more vulnerable to

cost increases due to carbon pricing and requirements to adopt expensive technologies for emissions reductions, and such policies have to be expected to have regressive distributional effects.¹ It has been well established that if climate policy is revenue raising, however, revenue can be redistributed to households (in a lump-sum per-capita fashion, e.g.) and that this revenue recycling is more than enough to compensate the regressive direct effects of carbon pricing and produce overall progressive effects. [Berry \(2019\)](#) shows this in the case of France for different parts of household energy consumption. It has also been found that besides the direct income on household purchasing power through increasing prices, there are also indirect effects through changes in productivity of regulated sectors and thus changes (drops, usually) in wages and capital rents. Commonly such indirect effects are found to be skewing policy impacts further towards progressivity, since high-income households derive a higher share of income from wages and capital rents while low-income households' income is made up of a higher share of transfers (see [Landis, Fredriksson, & Rausch, 2021](#); [Landis, Rausch, Kosch, & Böhringer, 2019](#); [Rausch, Metcalf, & Reilly, 2011](#), for some examples).

The second strand, also originating in studies by academic institutions, focuses on responsibility for global warming through the consumption of different parts of the global population. Several reports and studies explore the global and national inequalities related to this responsibility. On a global level Oxfam America's Climate Equality report ([Khalfan et al., 2023](#)) and the World Inequality Lab's Climate Inequality Report ([Chancel et al., 2023](#)), provide insights as to how responsibility for global warming and wealth/income are correlated. They conclude that the bulk of the responsibility lies with the rich part of the world and that lifting the poorest out of poverty at the current GHG intensity of consumption can easily be compensated by meaningful GHG emissions reduction by the richest part of the population. [Chancel et al. \(2023\)](#) present

¹A policy intervention is said to have a *regressive* impact, if its net cost relative to income is higher for low-income households than it is for high-income ones. The impacts are said to be *progressive* if it's the high-income households that incur higher costs relative to income.

139 a similar analysis but also provide numbers for inequalities *within* countries. They
 140 find similar patterns within countries as across the global population: The richest are
 141 responsible for far more global warming than their share in the population alone would
 142 justify. For Switzerland, Sotomo’s Energiewende-Index (Stückelberger, Bühler, & Her-
 143 mann, 2024) provides additional detail. Based on a consumer survey and the numbers
 144 from two CO₂ calculators, the report provides estimates of global warming caused by
 145 different parts of the population and it differentiates different consumption categories
 146 such as *base level*, *housing*, *mobility* (subdivided into flying and driving), *consump-*
 147 *tion*, and *food*.² As outlined in the first, introductory, paragraph, this type of analysis
 148 allows to make statements about responsibility-based obligations of different coun-
 149 tries to contribute to the global effort of reducing GHG emissions. This constitutes a
 150 valid input into international negotiations about climate policy, where different coun-
 151 tries coordinate on national emissions reduction targets and set individual policies to
 152 reach them.³ But when differences in responsibility for global emissions are analysed
 153 for households within the same country, it is less clear what conclusions can be drawn
 154 from that. The fact that also within countries, high-income households are responsi-
 155 ble for more global GHG emissions than are households with low incomes has earned
 156 some media attention and the general sentiment is that it must be ensured that rich
 157 households do their fair share when it comes to emissions reductions. But do they do
 158 that already under typical proposals for climate policy? And if not, what are policy
 159 instruments to mend that? Without administrative borders between households of dif-
 160 ferent income levels, it seems unlikely that policies will be differentiated by household
 161 income and the scope of designing national climate policies to target households of
 162 different income levels may be limited.

179 ²In German: “Grundverbrauch”, “Wohnen”, “Mobilität: Fahren”, “Mobilität: Fliegen”, “Konsum”, and
 180 “Ernährung”.

181 ³Note that this measure of responsibility for *concurrent* GHG emissions competes with other measures
 182 like historic responsibility for determining which countries should exert which efforts for combating global
 183 warming.

In this paper I shed light on both these distributional aspects by analysing the global warming potential (GWP) associated with the GHG emissions embodied in the consumption of different categories of goods across households of Switzerland. I exploit a data base of GWP estimates for consumption goods used in Household Budget Survey “Haushaltsbudgeterhebung” (HABE)⁴ provided by [Jakobs and Mutel \(2023\)](#) and couple it with the HABE data. By looking at GWP *per consumption good*, I can assess to what extent it may be possible to design climate policy to target goods that are consumed in particular by high income households or avoid other goods that are mainly consumed by low-income households. Comparing GWP embodied in households’ consumption patterns with their expenditure budgets, on the other hand, gives me a measure of their exposure to costly climate policy that encompasses all world regions and all GHGs.⁵

The contributions of this paper are twofold. First, it gives an overview over the extent to which policy makers can make sure that high-income households “do their fair share”. I find that for some consumption categories such as transport fuels, clothing, spare time activities, and eating out in restaurants high-income households consume a larger share of the national total than for other categories. Policies that cause emissions reductions for these goods (and luxury versions of those goods in particular) can help ensure that high-income households participate in the emissions reduction effort to an appropriate degree. Second, the paper discusses exposure of different income classes to costly climate policy. In that respect, my analysis lacks rigour and realistic policy detail compared to previous studies of EU and Swiss climate policy (e.g., [Landis et al., 2021, 2019](#)) but I provide an outlook on how consumption patterns influence distributional outcomes if climate policy extends in scope both spatial and in terms of GHG coverage. Fundamentally, distributional patterns can be expected

⁴Bundesamt für Statistik, Haushaltsbudgeterhebung (HABE) 2015–2017; HABE is a survey of almost 10’000 households representative of the Swiss population by the Federal Statistical Office (FSO).

⁵The analysis is precise for a “carbon” price that applies globally and to all GHGs according to their GWP and for a situation where production processes and supply chains have not yet adjusted to the increased cost of emitting. If other (globally uniform) policies are in place that make emitting activities more expensive, and as processes and supply chains adapt, my analysis loses in precision but I would claim that it is still indicative of general trends.

231 to remain similar as for the current regulation that concentrates on national emis-
232 sions of CO₂, since comparing emissions intensity of consumption between high and
233 low-income households looks qualitatively similar if we look at national CO₂ emis-
234 sions or global emissions of all GHG. If climate policy is carbon pricing, the per-capita
235 lump-sum redistribution to households will produce overall progressive outcomes in
236 the short to medium term.⁶ I find that including aviation, motor fuels, and trans-
237 port emissions in general in the CO₂-levy would make sense from the perspective of
238 distributing direct policy impacts. In either case, I can also report that the revenue
239 recycling mechanism of the CO₂-levy guarantees progressive outcomes.
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241 In the following, Section 2 describes the methodology and Section 3 shows results
242 for my analysis of inequalities in responsibility in climate change. Section 4 discusses
243 the possible implications of these findings for policy making. Section 5 summarizes my
244 findings and concludes.
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⁶In the long term, all emissions should be avoided and no carbon pricing revenue will be available for redistribution.

2 Data and method

In determining GWPs related with the consumption of different households, I follow and build upon previous work by [Jakobs and Mutel \(2023\)](#) which in turn implement methodology developed and described by [Froemelt, Dürrenmatt, and Hellweg \(2018\)](#). The different types of data and processing steps that are needed to arrive at numbers for per-capita GWP are described in the following and discussed in some more detail in [A](#).

Data and data processing

I use survey data from the 2015–2017 HABE,⁷ which gives information about income and spending for 9955 households (see [Oetliker & Siffert, 2022](#), for more information). HABE gives monthly spending in Swiss Francs (CHF) for a fine grained classification of consumption goods. In addition to values in CHF the survey records quantities purchased for food and beverages (in kg or litres) and for motor fuels (in litres). For several categories of vehicles, white goods, and electronic equipment, the survey also establishes the number of items owned per household.

GWPs of different consumption categories are given by [Jakobs and Mutel \(2023\)](#). [Jakobs and Mutel](#) combine life cycle assessment (LCA) data from the data bases ecoinvent 3.3,⁸ Agribalyse 1.2,⁹ and exiobase 2.2 ([Wood et al., 2015](#))¹⁰ to provide GWPs per HABE consumption category. [Froemelt et al. \(2018\)](#) provide a correspondence table of HABE consumption categories and the previously listed three LCA data bases in the supplementary material of their paper. For non-CO₂ greenhouse gases, GWP has been determined according to the Intergovernmental Panel on Climate Change (IPCC)’s 2021 methodology for calculating the warming equivalent to a given amount

⁷Bundesamt für Statistik, Haushaltsbudgeterhebung (HABE) 2015–2017

⁸<https://support.ecoinvent.org/ecoinvent-version-3.3>

⁹<https://doc.agribalyse.fr/documentation-en>

¹⁰<https://www.exiobase.eu/>

323 of CO₂ over a period of 100 years. This makes the unit for GWPs “kg CO₂ equivalents
324 (CO₂-eq)”.

326 The GWP data by [Jakobs and Mutel \(2023\)](#) are expressly constructed for use with
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328 HABE data: The consumption categories of the two data sets match exactly and most
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330 GWP factors can be multiplied by units of consumption (CHF, kg, litres, or number
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332 of devices owned) available in HABE.¹¹ Where units do not match, the HABE data
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334 insufficiently relates to consumed physical quantities and additional calculations need
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336 to be made. Since energy goods like heating fuels are bound to make up a relevant
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338 part of households’ GWPs, and since price differences across years and regions make
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340 spending in CHF an unreliable proxy for energy consumed, I convert CHF spent on
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342 fuels and electricity into energy consumed. For this, I take annual (and where possible
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344 regional) energy prices into account. For public transport, CHF spent on travel cards
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346 is a poor measure for distance travelled. To improve on HABE’s information that is
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348 given as spending in CHF, I employ data from the Mobility and Transport Microcensus
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350 ([Biedermann et al., 2017](#)). The micro-census is a survey conducted by the FSO asking
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352 Swiss residents about their mobility behaviour. I use km travelled on different modes
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354 of public transport for different household types from the micro-census and compare
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356 them with CHF spent according to HABE and derive a transport mode and household
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358 type specific conversion factor in km per CHF.

356 Categorizing households

358 To analyse which part of the Swiss population is responsible for how much GWP
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360 through their consumption, the set of households in the data from the HABE survey
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362 can be partitioned into different groups. HABE records a diverse set of socio-economic
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364 indicators that describe the entire household or its representative or household head
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366 (i.e., the household member that makes the largest contribution to total household

367 ¹¹GWP are estimates rather than precise numbers and [Jakobs and Mutel \(2023\)](#) provide mean, median,
368 and standard deviation of the distribution of estimated GWP. For the purpose of this analysis, I use the
mean estimates.

income). Indicators describe properties such as “household size”, “degree of urbanization”, “ownership of residence” and many more. An important dimension along which GWP is discussed in the literature is income. In order to be able to control the size of the income groups I create my own partition into income quantiles. To this end, I need to define an appropriate measure of income and order households according to it. I observe that current (annual or monthly) income may fluctuate for some households due to special circumstances which households can compensate through intertemporal consumption smoothing. I thus view current consumption expenditures as a more reliable measure of how well off households are (of their lifetime income) than current income. Assuming that income is shared equally within a household, total income not only needs to be divided by the size of the household (giving per-capita income), but it also needs to be considered that the purchased consumption goods (such as electronic devices, lighting, or room heat) are shared, and consumption is more efficient at generating consumer utility from a given amount of expenditure for big households than for small ones. The Organization for Economic Co-operation and Development (OECD) established a de facto standard for equivalence scales to compensate for this. This so-called “modified OECD equivalence scale”¹² measures the first adult in a household with weight 1 and additional adults with weight 0.5 and additional children with weight 0.3. Dividing total household income with this equivalence scale gives equivalent income.

Combining the above two considerations, I group households into income quantile groups¹³ by ordering them according to their “equivalent lifetime income” which is proxied by dividing current household expenditures by the equivalence scale.

¹²See, e.g., https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Equivalent_income

¹³Quantile groups of (equivalent lifetime) income are constructed by ordering households according to income and grouping together those that are situated between quantiles of income. For 10-quantiles, or deciles, this means that the first decile group corresponds to the 10 percent of the population with the lowest (equivalent lifetime) income (their income is between 0 and the first decile of income). The second decile group comprises households with income between the first and second income decile, etc.

3 Responsibility for global warming

For the purpose of this study, a household's responsibility for global warming is determined by the GHG emissions related to the production and consumption of goods and services consumed by household members. This includes emissions from all GHGs and emissions caused across the globe. This responsibility is established by considering *current* annual consumption and thus accounts for *current* GHG emissions.¹⁴ My measure of responsibility therefore counts the contribution of consumers towards current global GHG emissions, a number that needs to go to (net-)zero if we want to stop global warming at a level that is sustainable. This can inform policy making in two ways. First, policies must ensure that emissions from all parts of the population and all important categories of consumption are reduced. An account of what parts of the population contribute to global warming through which categories of consumption can help assess if current or proposed policies do a good job at this. Second, policy frameworks should make sure that those who need to change their consumption patterns also have the means and incentives to do this. To this end, I show in the following how both GWP per person and GWP per CHF expenditure is distributed across different parts of the Swiss population as well as how different consumption categories contribute to the overall GWP.

3.1 GWP for different households

For establishing the responsibility for global warming on a per-person basis, the GWP from different consumption categories for a household are added up and divided by the number of persons in that household. By grouping households according to their socioeconomic properties, averages for different household groups can be established and the typical GWP can be compared across the groups. At the same time, heterogeneity within such groups turns out to be large, and therefore, this paper shows not only mean

¹⁴That is the emissions from this year or some recent year in which the consumption goods in question were produced.

GWP per population group but summarizes the distribution of observations within those groups using box plots.¹⁵ The boxes drawn in these plots are characterized by three horizontal lines: the y-axis values for the observations at the first, the second, and the third quartile. The boxes are complemented by whiskers that reach out to the last observation that lies less than 1.5 times the interquartile range (the distance between the first and the third quartile) beyond the box. It is customary to plot outliers beyond the whiskers as explicit dots, but I abstain from showing them here, directing the attention to the main bulk of the ‘observations rather than to single observations. Superimposed on the box plots are the weighted means of the y-axis values within the respective household groups (depicted by an empty circle).

I find (equivalent) lifetime income to be the most important criterion for determining GWP. To see this, consider that lifetime income is strongly related to household expenditures. Figure 1 shows that, some considerable heterogeneity within income decile groups notwithstanding, per-capita expenditures increase notably for higher deciles of lifetime income. At the same time, emissions-intensity of consumption (measured as GWP per spending in kg CO₂-eq per CHF) is higher for low-income decile groups than for high-income ones (top panel in Figure 2), but this trend is weaker and the trend in per-capita spending dominates the overall trend in per-capita GWP shown in Figure 3 (top panel). At the same time, within-decile group variation in results is bigger for per-capita GWP than for per-capita spending, driven in part by the considerable within-decile group variation of GWP *intensity* shown in Figure 2.

¹⁵See Krzywinski and Altman (2014) for a more in-depth discussion of box plots.

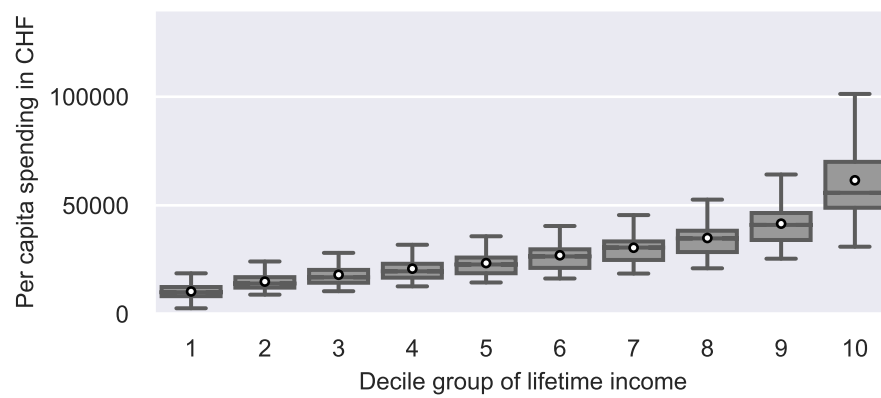


Fig. 1: Annual per-capita expenditure within and across decile groups.

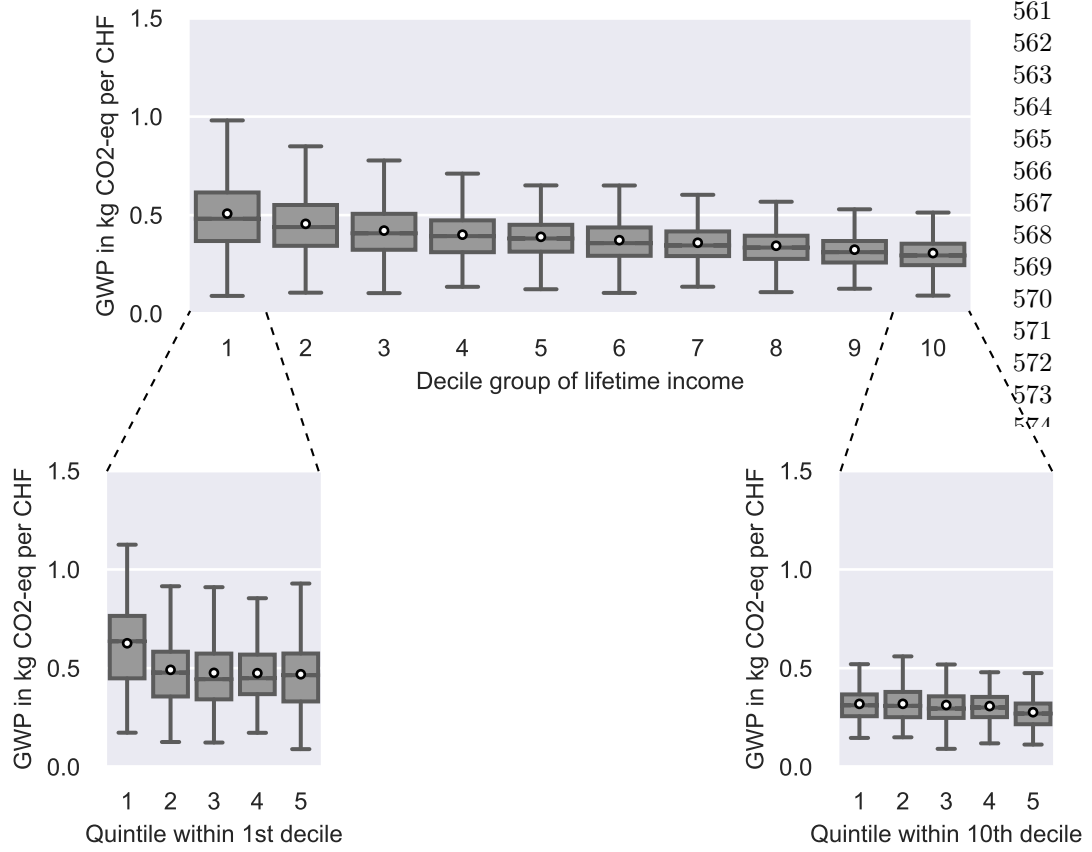


Fig. 2: GWP intensity in kg CO₂-eq per CHF within and across decile groups (top panel) and within the first and tenth decile groups (bottom panels).

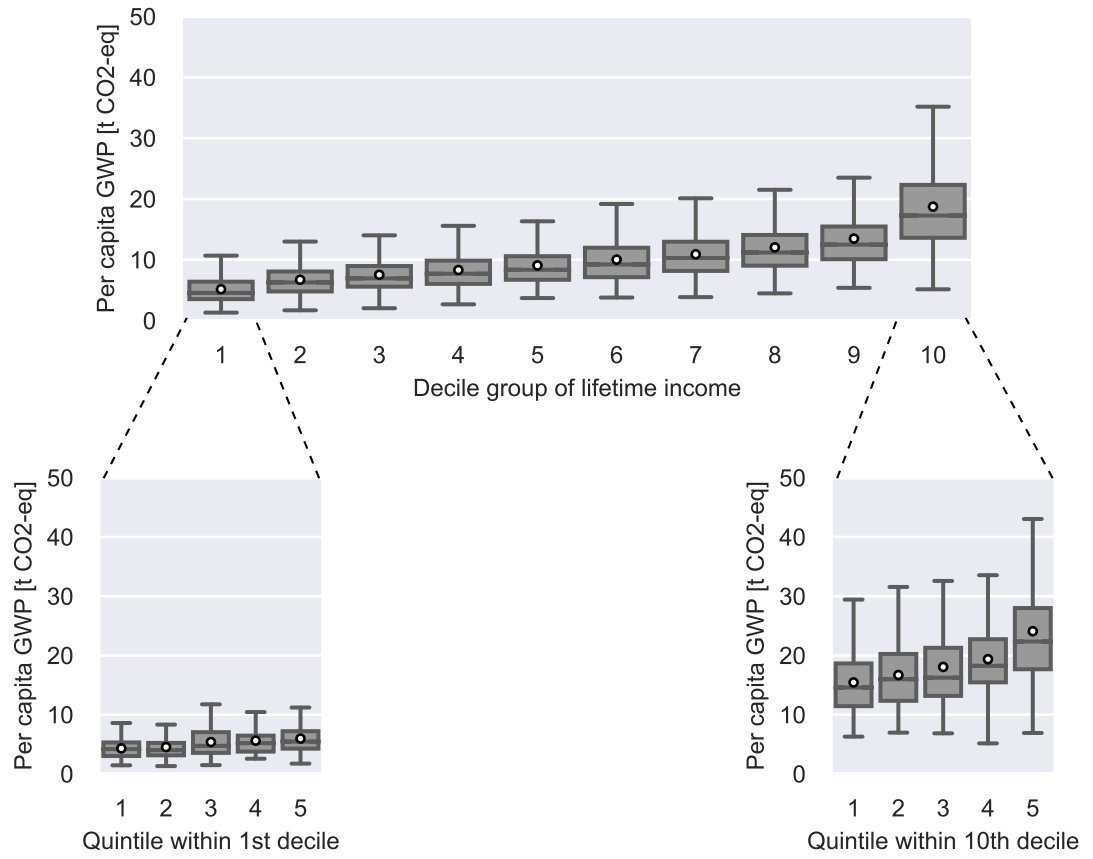


Fig. 3: Per-capita GWP within and across decile groups (top panel) and within the first and tenth decile groups (bottom panels).

While we can identify clear trends of the means across decile groups of the population in the top panels of Figures 2 and 3, further subdividing decile groups into 2-percentile-groups (or quintile groups within decile groups; see bottom panels) does not add as much additional insight. In Figure 2 there is no clear and significant trend visible for GWP intensity within the tenth decile group. The figure for the first decile group suggests that the negative correlation between lifetime income and GWP intensity may also be at play within the decile group but the numbers of observations within decile sub-groups is not high enough to say this with confidence. (A more detailed discussion of this can be found in C). What we can say, is that the first 2-percentile-group has a higher mean GWP intensity than the following four. Similarly, in Figure 3, no clear trend is visible across 2-percentile groups within the first decile group, and a positive correlation between lifetime income and per-capita GWP suggests itself within the tenth decile group, with the highest-income 2-percentile-group showing significantly more per-capita GWP than the preceding four.

Among the other socio-economic descriptors along which the population can be divided in HABE, I find household composition, home owner status, and the degree of urbanization to be the ones that showed the most meaningful differences in per-capita GWP.¹⁶ Figure 4 gives an overview of the results for these groupings of Swiss households. The top panel show that the household composition (in terms of number of household members and their age) is another relevant determinant of average per-capita GWP. If household heads are retired, the household's per-capita GWP is noticeably lower, and big households can use their economy of scale to live with lower per-capita GWP. The bottom panel shows that home owners, on average, emit more per capita than do residents who rent their home (left) and that residents of peri-urban areas (the "Agglomeration") have the the highest and the residents of urban areas the lowest per-capita GWP (even though this trend is not very distinctive).

¹⁶Besides household composition, home owner status, and the degree of urbanization, I checked gender of household head, canton of residence, region of residence, and pensioner status of household head.

691 The fact that the correlation of lifetime income and per-capita GWP is strong,
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693 and that households with different household composition, home owner status, and
694 degree of urbanization will also have different income, begs the question if different
695 GWP can just be explained by the income differences alone. The analysis in [B](#) shows
696 that that differences in GWP across those three household properties persist if we
697 differentiate them within decile groups of lifetime income. This suggests that these
698 different household types have different per-capita GWP due to differing consumption
699 patterns: They allocate their income in different ways across the consumption goods
700 they purchase. This is be further analyzed in the following.
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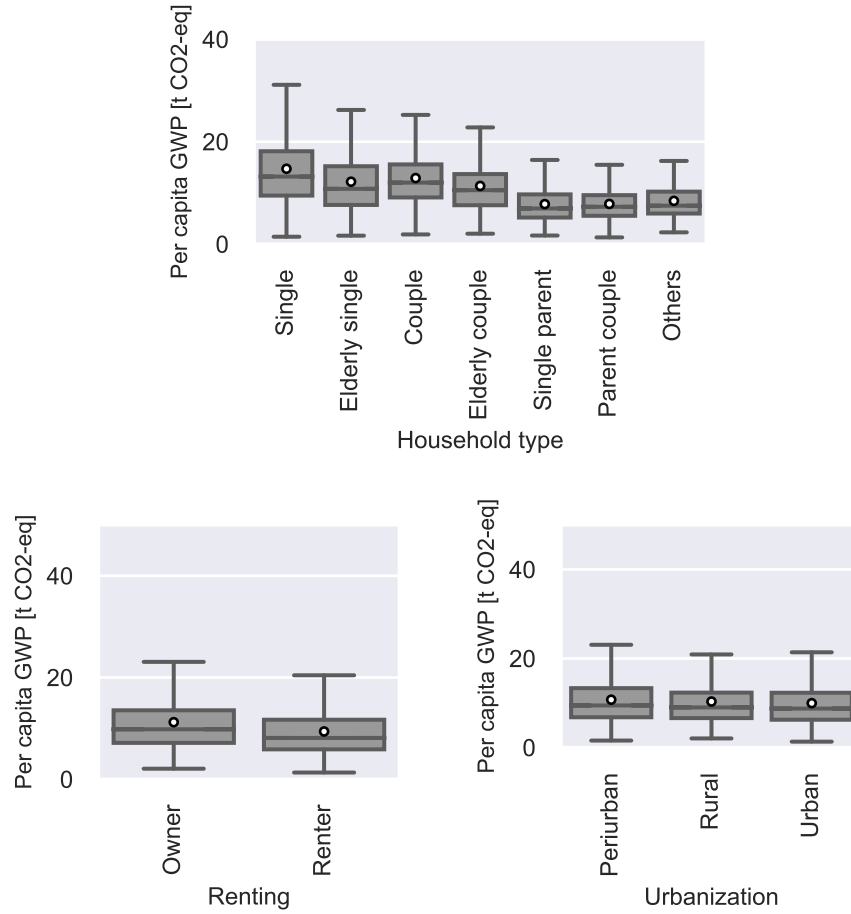


Fig. 4: GWP within and across different sets of household groups. Top panel: Households grouped by household composition. Households are labelled “elderly” if one household member is aged 65 or older. Bottom panel: Households grouped by ownership of dwelling (left) and degree of urbanization (right).

783 3.2 GWP by category

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785 Figure 5 shows how mean per-capita GWP is composed of contributions from different
786 consumption categories. We see that the main contributors to GWP across decile groups
787 are the four categories “food and drink”, heating (“Central and district heating”
788 plus “Gas and other heating fuels”), “private transport” (with the main component
789 of gasoline and diesel consumption), and “electronic equipment”. For households in
790 decile groups of higher affluence, the emissions from these categories tend to increase.
791 It can be noted that emissions from food, heating, and electronic equipment correlate
792 less strongly with income than does private transport. For categories of consumption
793 other than the four above mentioned, emissions make up a minuscule part in the
794 first decile group but increase to relevant size for the more affluent decile groups.
795 These consumption categories include goods that pertain to restauration and lodging,
796 clothing and footwear, furniture and household equipment, hobbies, holidays, and
797 education.

800 I note that a (costly) reduction of the emissions from the latter group of goods
801 would not impact the lowest-income decile groups much but would on its own also not
802 come close to reaching a net-zero emissions target. Without avoiding the emissions
803 from “necessary” consumption goods like food and drink, heating, private transport,
804 and electronic equipment, a net zero emissions world seems impossible.

805 Figure 6 shows per-capita GWP from consumption related emissions for different
806 household compositions in the top panel. Household composition is determined by the
807 number of household members, their age, and their family status. The results show
808 two main trends: First, households with elderly people are (on average) responsible for
809 fewer emissions mainly because they use less transport, but their emissions from heat-
810 ing are somewhat higher. Second, the bigger the household, the lower are the per-capita
811 emissions. They exhibit economies of scale for both transport and heating: Household
812 members can share both cars and heated homes in order to enjoy a given level of
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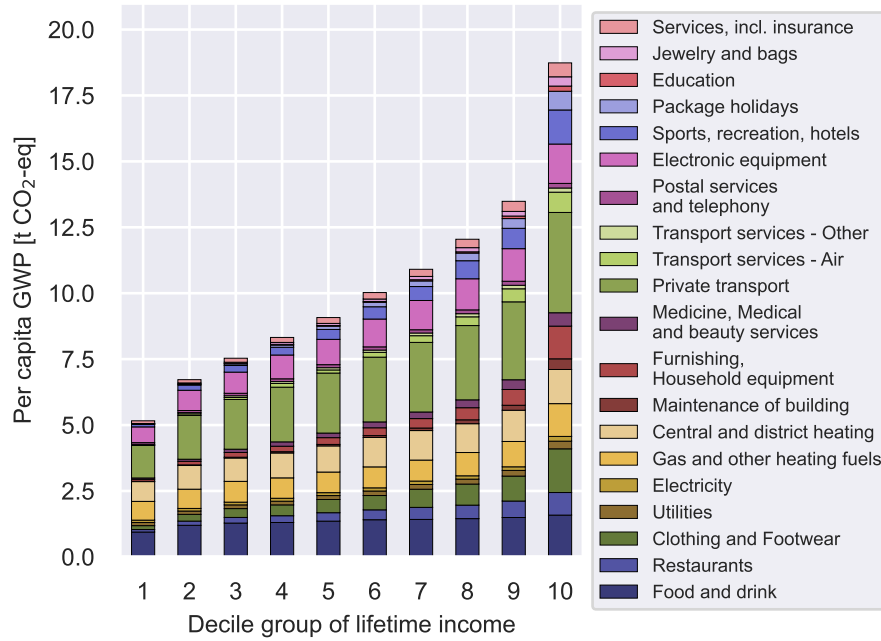


Fig. 5: Annual per capita GWP from consumption across expenditure decile groups.

energy services at lower per-capita expenses and emissions. An exemption is transport with the elderly. Elderly couples cause more per-capita emissions through transport than do elderly singles. The bottom panel of Figure 6 confirms that in per-household terms, big households do emit more for both transport and heating (compare “Single” with “Couple” households and “Elderly single” with “Elderly couple” households).

Figure 7 shows how per-capita GWP across categories depends on home ownership (top panel) and degree of urbanization (bottom panel). As seen in Figure 4 before, home owners, on average, emit more per capita than do people who rent their dwelling. While other consumption categories contribute to the difference as well, a big part of it is coming from emissions in heating. Home owners heat less with district and central heating and create more emissions through heating overall. Note that the HABE data does not allow for a conclusive answer about what causes these higher emissions

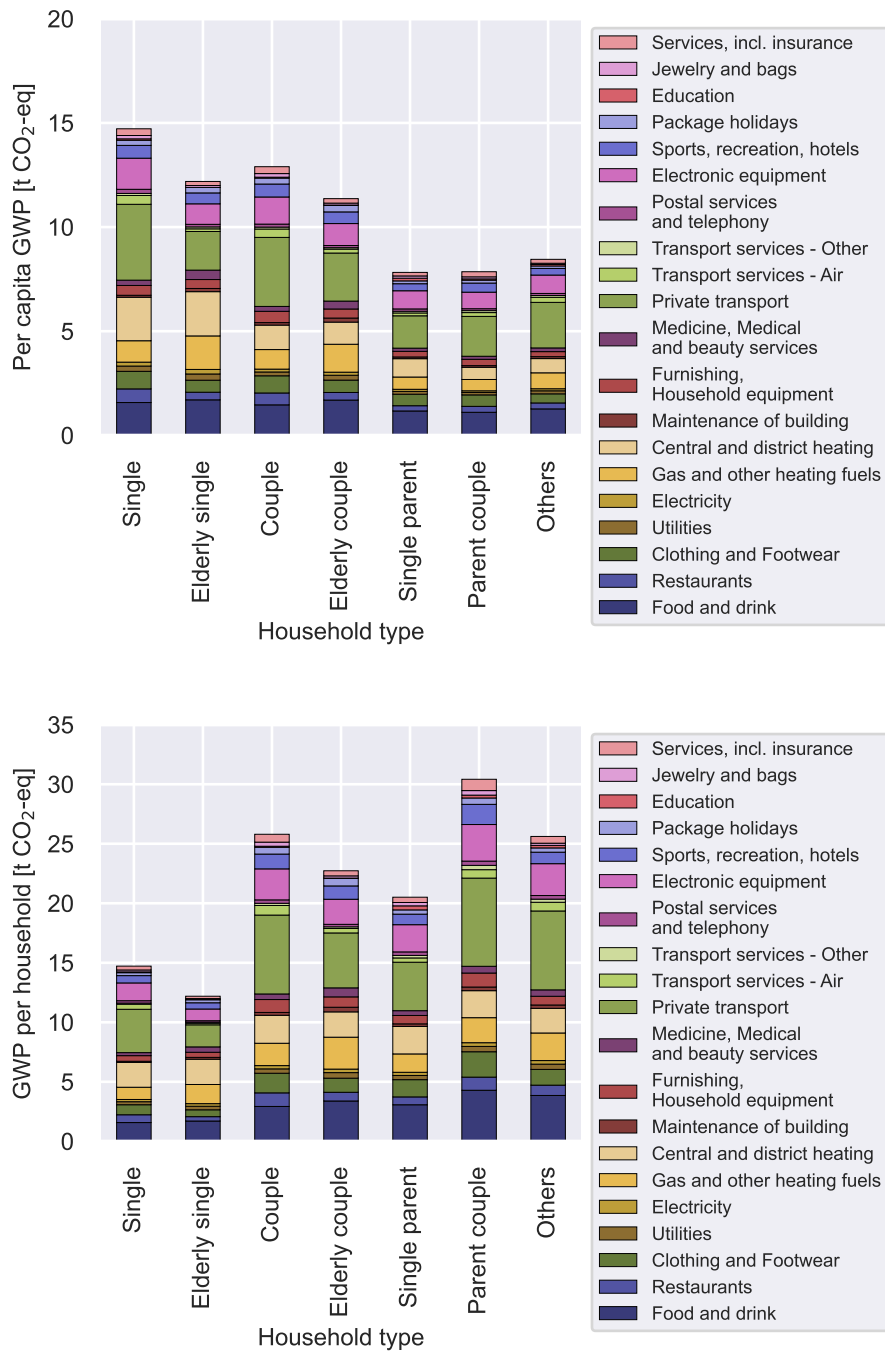


Fig. 6: Annual GWP from consumption across different household types. Per-capita GWP is given in the top panel and per-household GWP in the lower panel. Elderly singles are aged 65 or older, elderly couples are couples where at least one person is aged 65 or older.

because heating demand depends, among others, on the floor area that requires heating
and on the quality of insulation, factors that are not captured by HABE.

The degree of urbanization is also predicting per-capita emissions in different
households to a visible degree (lower panel of Figure 7). Here, it is transport that is
the main driver of these differences. Households living in peri-urban settings have the
highest per-capita emissions from transport, while households living in urban areas
have the lowest. The same holds for overall per-capita GWP.

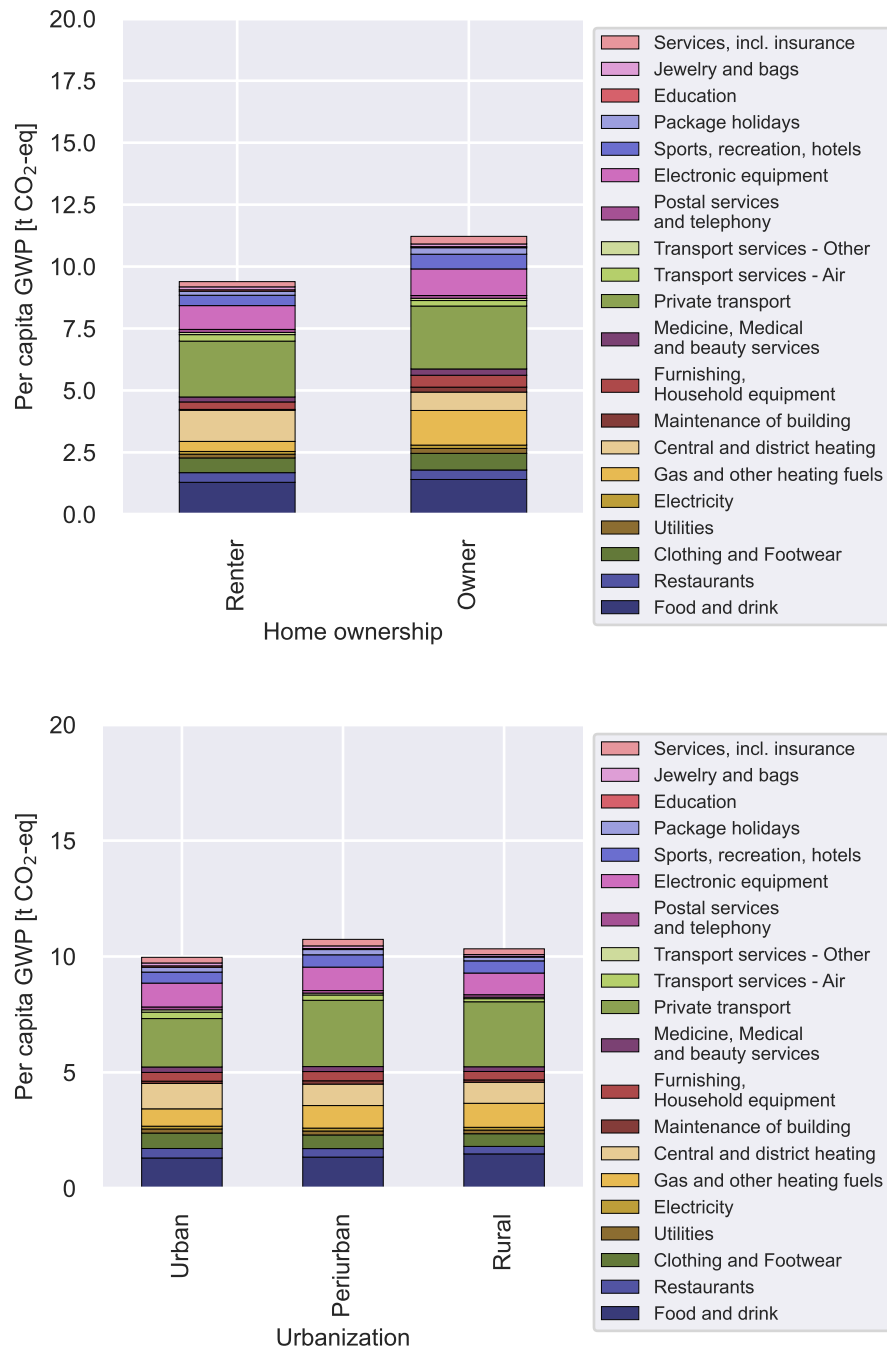


Fig. 7: Annual per capita GWP from consumption across different household types. Households are differentiated by ownership of dwelling (top panel) or by degree of urbanization (bottom panel).

3.3 Transport and heating

Transport and heating are two of the most important categories of consumption in terms of GWP and their emissions arise mostly within the borders of Switzerland. They are thus central to Swiss climate policy targets and deserve closer examination. Figures 8 and 9 in their top panels show excerpts of Figure 5 for the two broad consumption categories transport and heating and subdivide the two into the most disaggregated categories that HABE provides. Two things become apparent.

First, different categories of transport contribute to overall GWP to different degrees. The fine-grained differentiation of transport modes and fuels (see Figure 8) reveals that GHG emissions from transport activities are dominated by fuel use for private transport. Other categories that matter are air transport and emissions from package holidays (but note that, besides transport, this category also includes emissions from heating hotels, etc.). The different modes of public transport account for a much smaller share of GHG emissions in Switzerland, since they are mostly driven by low-emissions electric power. The expenditure data from HABE does not allow for a very fine grained analysis of different heating fuels but much rather groups heating expenditures into direct payments for heating fuels (mostly by owners of dwellings) and payments for central and district heating (mostly by households who rent).

Second, emissions (and thus exposure to policies requiring costly emissions reduction) rise much slower with income for heating than they do for transport. The bottom panels of Figures 8 and 9 emphasize this point by displaying GWP-intensity (emissions relative to overall household spending) for different decile groups. For both categories, low-income households emit more GHGs per CHF (and are thus likely to incur higher costs from reducing emissions relative to their overall consumption budget) than do high-income households. But the trend is much stronger for heating than it is for transport and is even more extreme for air transport and package holidays.

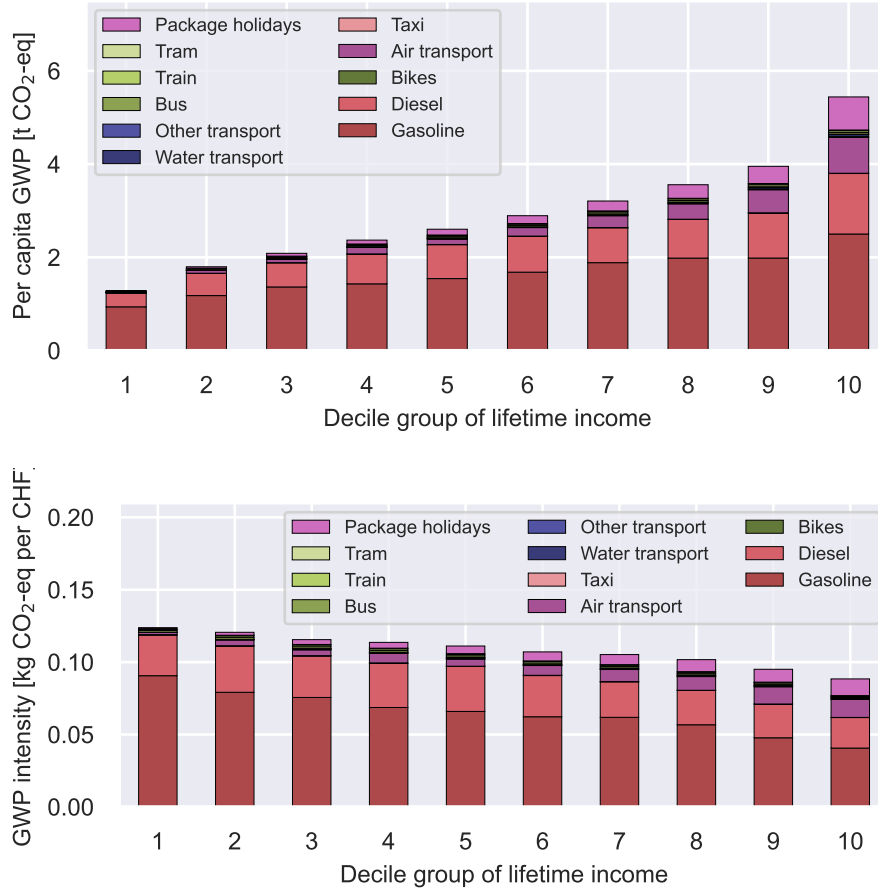


Fig. 8: GWP related to transport across expenditure decile groups. The top panel shows annual per capita GWP, the bottom panel shows transport related GWP intensity of household expenditure.

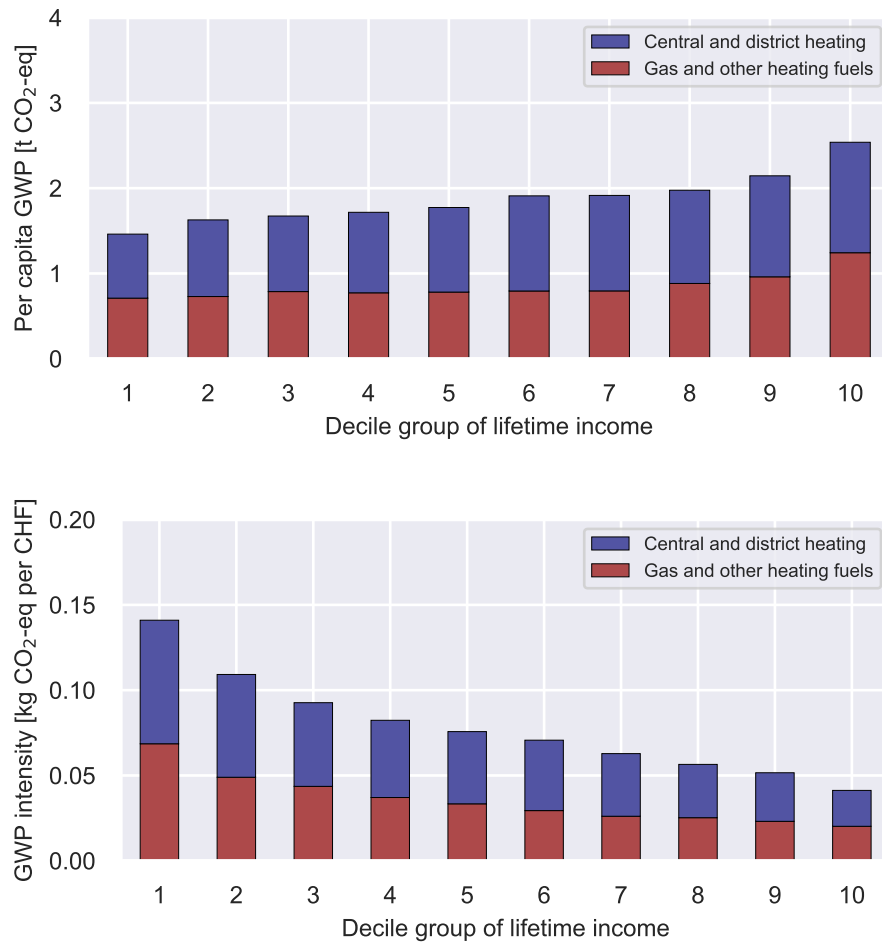


Fig. 9: GWP related to heating across expenditure decile groups. The top panel shows annual per capita GWP, the bottom panel shows heating related GWP intensity of household expenditure.

1151 3.4 Sensitivity with respect to aviation emissions and other 1152 1153 uncertainties 1154

1155 The GWP estimates by [Jakobs and Mutel \(2023\)](#) use information about GHG emis-
1156 sions caused by the consumption of different consumption goods and compares their
1157 “effectiveness in causing radiative forcing” to that of emitting CO₂. This comes with
1158 two drawbacks. First, consumption goods are related to emissions of different GHGs
1159 in a way that is specific to Switzerland but does not differentiate further according to
1160 household types, place of residence or income. This works well for consumption goods
1161 like single food items where quantities are given in terms of kilograms or litres by
1162 HABE. But with broader consumption categories like meals in *restaurants* where con-
1163 sumption is given in terms of CHF by HABE, we have little information about how the
1164 consumption of different households is composed. Both the amount of calories con-
1165 sumed per CHF as well as the composition of such meals in terms of meat or vegetarian
1166 options may well depend on the income level of the consumers. I have similar reserva-
1167 tions against *clothing and footwear*, where spending in CHF does neither reveal what
1168 quantities of textiles have been purchased nor their quality (leather, cotton, synthetic
1169 textiles). For both *restaurant* visits and *clothing and footwear*, my analysis suggest
1170 a distinctive increase in GWP from per-capita consumption across income levels. To
1171 the extent that high income individuals purchase more expensive versions of the com-
1172 modities (and thus smaller amounts of physical quantities per CHF), this effect may
1173 be more attenuated in reality. Without more detailed data about what specific quan-
1174 tities are purchased by different households within these consumption categories little
1175 can be done to alleviate this data problem.

1176
1177 Second, the GWP methodology treats given GHGs the same, independently from
1178 the circumstances in which they have been emitted. Yet in aviation, the consequences
1179 of aeroplanes’ emissions for the climate depend crucially on flight altitude, geography,
1180 atmospheric conditions and time of day ([Frömming et al., 2021](#)). Thus, conditions
1181
1182

along flight routes result in radiative forcing/global warming that is higher than GWP would suggest. It is a commonly applied method to multiply the GWP with a radiative forcing index (RFI) factor to account for this difference for *average* flights. Jungbluth and Meili (2019) summarize the literature and find that different studies use RFI factors of 1 to 2.7 and recommend a factor of 2.

To see how my results change if I take these insights into account, I multiply the aviation-related GWP of Jakobs and Mutel (2023) by the recommended RFI factor of 2 in Figure 10. Since plane tickets are not only booked directly but also as part of package holidays, I give the same treatment to the consumption category *package holidays*. This assumes that the majority of GWP in package holidays is caused by flight travel and the revised GWP for *package holidays* thus constitutes an upper bound. It becomes apparent (Figure 10), that the two consumption categories, the expenditures for which are distributed progressively (see bottom panel of Fig. 8) are more important for overall GWP from consumption than my previous analysis (Fig. 5) suggested. This emphasizes the importance of addressing emissions in aviation in climate policy.

When I consider how exposed different households are to costly climate policy in the following section 4, I do not include this RFI factor. I assume that the effects of non-CO₂ emissions on radiative forcing can be reduced cost effectively by re-routing plane routes and that pricing emissions will increase the cost of flying commensurate to the GWP displayed in Figures 5–8, i.e. as derived from the GWP estimates given by Jakobs and Mutel (2023).

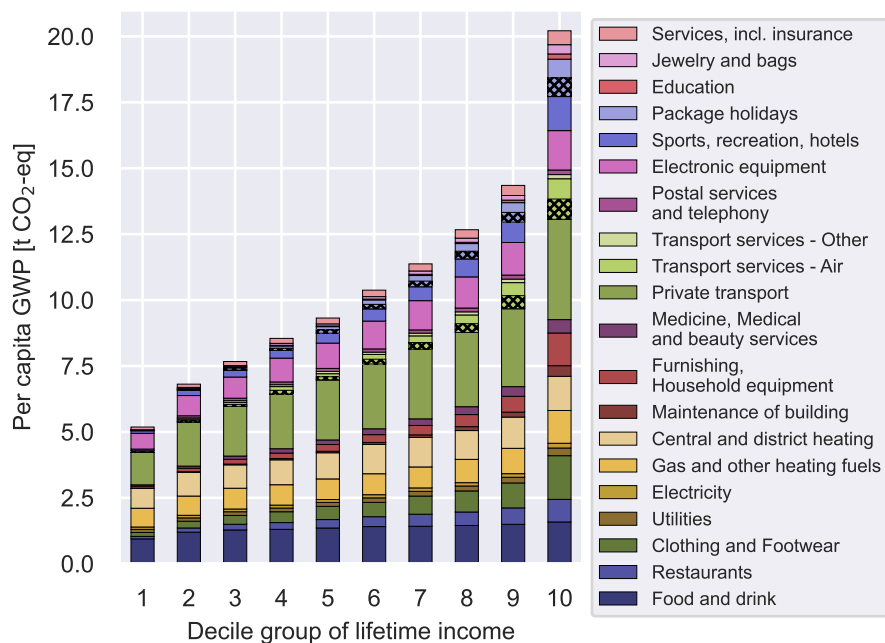


Fig. 10: Categorization of GWP per capita if emissions from aviation are multiplied by an RFI factor of 2. The *additional* GWP from this adjustment is highlighted by a chequered pattern.

4 Climate policy design	1289
	1290
The analysis above shows that different households have different responsibilities for	1291
consumption-based GHG emissions, and that emissions from some consumption cate-	1292
gories such as airborne passenger transport are mostly generated by the more affluent.	1293
	1294
Yet, the most important consumption categories for overall emissions are food, heat-	1295
ing, transport and electronics, which make up significant shares in overall spending	1296
for low-income households in particular. This highlights the fact that we cannot get	1297
around increasing the cost of consumption for low-income households if Switzerland's	1298
net-zero emissions target is to be met (I assume that emission free technologies for	1299
producing different consumption goods are more expensive than their polluting coun-	1300
terparts at least in the short and medium run and that the state will not fully take	1301
over these additional costs).	1302
	1303
In the following, I explore some options for design of climate policy that allow pro-	1304
tecting low-income households from disproportionate policy cost and make sure that	1305
high-income households do their fair share. The reasoning is that low-income spend	1306
their income mostly on necessities and are hurt more than high-income households	1307
by increasing costs of consumption and that low-income households may be credit-	1308
constrained and have difficulties to pay high up-front costs of investments for green	1309
(that is, non-GHG-emitting) technologies that may be even cost-saving in the long-run.	1310
	1311
Another reason for "letting the rich go first" is that technology costs may decrease	1312
over time, and that if those who can better afford it invest first, the cost of invest-	1313
ments in green technologies may be lower when other, less affluent households make	1314
these investments later on.	1315
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1335 4.1 Regulating for emissions reduction by high-income

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1337 households

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1339 The analysis of emissions by consumption category and income decile group lets me
1340 identify several consumption goods the emissions of which could be regulated with-
1341 out affecting the consumption of low-income households too much. Examples for these
1342 appear to be air transport, lodging in hotels, and restaurant visits.¹⁷ Policies that
1343 target such consumption goods specifically are almost certainly costing low-income
1344 households very little. But at the same time, emissions related to these consump-
1345 tion goods make up only a limited share of overall emissions that Swiss consumers
1346 are responsible for, and it is imperative that emissions from other, less conveniently
1347 distributed consumption goods be reduced.

1354 When regulating emissions from consumption categories that also make up a com-
1355 parably big share in expenditures of low-income households (such as food, transport,
1356 heating, and electronics), policy makers can still try to design policies such that high-
1357 income households are incentivized to invest first, while green technologies are still
1358 expensive, and low-income households profit from lower cost established green tech-
1359 nologies later. In markets where low-emission technologies are seen as a luxury good,
1360 this may to some extent be the natural market outcome. For battery electric vehic-
1361 less (BEVs) it is argued that the market introduction worked smoothest, when Tesla
1362 initially catered to a niche market with a highly specialized sports car and only later
1363 reduced cost and provided more affordable cars to a wider audience. But policy may
1364 push this tendency even further by identifying *luxury* versions of given goods and reg-
1365 ulating these more stringently than the non-luxury versions. This works best if the
1366 luxury version of a good is also more emissions-intensive. To remain with the example

1376 ¹⁷In the case of restauration, where consumption is measured in CHF it is implausible that high-income
1377 households consume the same amount of calories per CHF as do low-income households. We have to expect
1378 that – contrary to my modelling assumptions – emissions per CHF spent in restaurants may be higher
1379 for low-income households than for high-income ones. My findings for how emissions from eating out in
1380 restaurants are distributed across income may therefore be biased towards overstating emissions for high-
income and/or understating emissions for low-income households.

of cars, climate policy could aim to make non-CO₂-emitting technologies mandatory for large cars such as sports utility vehicles (SUVs) or two seated sports cars. Alternatively, fleet standards for CO₂ emissions per kilometre could be complemented by a maximum amount of CO₂ per kilometre *any* model within the fleet may emit (with potential exemptions for family cars with more than five seats).

But, admittedly, if we move away from transport the identification of luxury versions of goods becomes more difficult and regulating them may become more controversial. If we take the size of a dwelling (divided by the number of occupants, e.g.) as a measure of luxuriousness in the context of heating systems, and a retired couple lives in a spacious house they own, should they be forced to invest in renewable heating system even if they have difficulty affording the necessary investment from their savings? Or do we only demand a renewable energy heating system when the property changes hands?

To summarize, singling out luxury goods and focusing emissions reduction efforts on them may be possible in some cases but these make up a limited share of overall emissions and if Switzerland wants to reach net-zero emissions, GHG emissions reduction efforts will also have to include transport and heating which will necessarily affect low-income households as well. The following section discusses to what extent low-income households may be impacted disproportionately by policy costs and makes some suggestions for policy designs to avoid this.

4.2 Protecting low-income households from disproportionate policy costs

Pricing CO₂ emissions in Switzerland

Climate policies at our disposal can generally be categorized in three types: carbon pricing, mandates for clean technologies, and subsidies for investments (or research)

1427 into clean technologies. Carbon pricing is appreciated by many economists as a cost-
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1429 efficient measure since all economic actors and markets see the carbon price and can
1430 use the information at their disposal for finding low-cost options to reduce emissions
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1432 (Boyce, 2018). Government mandates and subsidies on the other hand rely on the
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1434 government to identify the best way for reducing emissions and run the risk of being
1435 much more expensive since the state does not have all the information about options
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1437 for emissions reduction that the different actors in the different markets have. Landis
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1439 et al. (2019), e.g., show that for policy design similar to the current Swiss policy
1440 framework, carbon pricing policies can be expected to keep overall costs of achieving
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1442 emissions reduction significantly lower than policy packages that focus on mandating
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1444 and subsidizing specific technologies alone. This efficiency argument and the fact that
1445 carbon pricing is an established policy instrument in climate policy (see, e.g., the
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1447 emissions trading systems for energy-intensive industries in Switzerland and the EU,
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1449 the CO₂-levy on heating fuels in Switzerland, and the recent expansion of emissions
1450 trading to transport and heating fuels [termed ETS 2] in the EU) makes it plausible
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1452 that it will remain one of the corner stones of Swiss climate policy.

1453 At the same time it is the policy that creates the highest direct policy cost to
1454 consumers even in the short term: Not only are consumers paying for more expensive
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1456 green technologies if they reduce emissions, but they also pay the CO₂-levy on the
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1458 emissions they have not yet avoided through such measures. And, again, the obser-
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1460 vation that the consumption of low-income households is on average associated with
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1462 the highest emission intensity of consumption (their consumption causes more emis-
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1464 sions on a per CHF-spent basis than higher-income ones, see Figure 2) suggests that
1465 the direct costs incurred from carbon pricing tends to impact low-income households
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1467 most relative to their consumption budget.

1468 This observation about direct policy cost has to be qualified by the expected effects
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1470 of two additional mechanisms (indirect effects). One is driven by what happens in
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industries and markets when climate policy makes producers reduce emissions themselves. The consequence of costly emissions reduction in productive sectors is that real wages and capital rents decrease and this affects high-income households more than it does low-income ones. The second mechanism is part of carbon pricing itself and is subject to policy design choices: Carbon pricing generates revenues that can be employed in different ways and one way or another these revenues are recycled back to the economy, which can have its own distributional consequences.

For existing policies, the two mechanisms have been discussed in the academic literature at some length. For the indirect policy cost via reductions in wages and capital rents, the findings generally indicate that the regressivity¹⁸ of the direct effects of carbon pricing on consumer prices is much reduced if not neutralized by the indirect effects. [Rausch et al. \(2011\)](#) provide a transparent analysis and compare the direct (uses side) and indirect (sources side) effects on the distributional outcome of carbon pricing in the case of the US. But while the regressive effects on consumer prices are (mostly) neutralized *on average*, the wide heterogeneity in effects on low-income households¹⁹ means that the most gravely impacted households are still among the low-income decile group ([Landis, 2019](#); [Landis et al., 2019](#)). This highlights the necessity to further think about the distributional effects of how the revenue from carbon pricing is recycled. When I compare different ways of recycling carbon pricing revenue in a previous study ([Landis, 2019](#)), I find that lump-sum per-capita recycling (i.e., giving each Swiss resident the same amount of money; this is what is currently being done with two thirds of the revenue from the CO₂-levy) gives good results. This finding can by and large be explained by the fact that for low-income households, the lump-sum transfer is on average larger than their outlays on the CO₂-levy, while the direct costs of the CO₂-levy exceed the lump-sum transfer in the case of high-income households.

¹⁸A policy or an aspect of policy design is called regressive, if its cost to low-income households compared to their expenditure budget is higher than the cost to high-income households *on average*.

¹⁹The wide heterogeneity of emission intensity in Figure 2 shows main the source of this heterogeneity of outcomes.

Figure 11 shows the resulting impacts of carbon price based Swiss climate policy on consumer utility in 2035 and 2050 and their distribution across households in income quintile groups. It becomes evident that relative consumption utility impacts on low-income households, *on average*, are smaller than those on high-income households. Yet, the households with the worst utility impacts are still to be found in the lower-income quintile groups, even with revenue recycling in place.

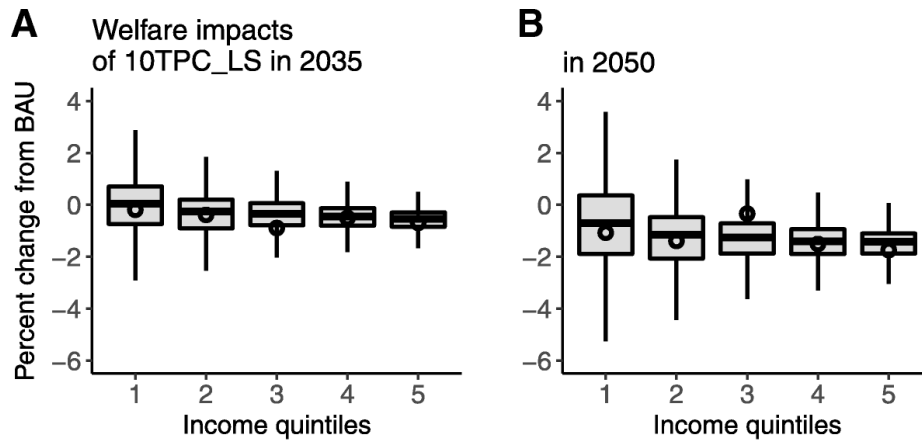


Fig. 11: Welfare costs of carbon price based climate policy for Switzerland in 2035 (panel A, left) and 2050 (panel B, right) according to Landis (2019). Two thirds of carbon pricing revenue is recycled back to households in a per-capita lump-sum fashion. “Welfare impacts” of climate policy reflect how much more or less consumption utility households are able to achieve relative to the consumption budget in the base-line (equivalent variation relative to baseline expenditure budget).

Source: Landis (2019)

Pricing ghg emissions globally

The findings above are for pricing of CO₂ emissions in Switzerland. But as time goes by, the scope of climate policy should expand (to other GHGs as well as to other regions). In the following, the findings in Section 3 shall be used for illustrating that similar patterns can be expected to apply if climate policy encompasses all GHGs from all around the globe. From the point of view of Swiss consumers, this may occur, e.g.,

if all GHG emissions in Switzerland are regulated using the same or similar emissions prices and imports are taxed at the borders for their embodied emissions.²⁰

I restrict my analysis to the short term. That means after introduction of emissions pricing, firms do not adjust production processes, and all they can do is increase prices according to the emissions associated with their production. Equally, households do not adjust consumption patterns but just face higher costs with constant expenditure budgets.²¹ Without such adjustments, the additional costs of consumption goods from emissions pricing is the product of the consumption goods' emissions content and the emissions price. For exploring the qualitative distribution of impacts of a price on global GHG for Swiss consumption, I assume that emissions are priced at CHF 100 per tCO₂-eq.

The upper panel of Figure 12 shows the mean direct cost of such an emissions price on households in the ten decile groups (blue solid line). As low-income households have smaller expenditure budgets the amount of emissions related to their consumption is small as is their direct cost from emissions pricing compared to high-income households. I assume that the revenue from this is collected by the government and fully redistributed to households in a per-capita lump-sum fashion. The resulting additional refunds for households are shown in Figure 12 as the dashed red line. (The fact that this line is not perfectly horizontal is due to the fact that it shows the averages of *per-household* revenues and that average household size varies slightly across decile groups.) It becomes apparent that low-income households pay less for carbon pricing

²⁰European climate policy is currently moving into that direction. Recently, the EU has decided to regulate emissions of motor and heating fuels with a second emissions trading system, thus moving closer to price based regulation of all domestic GHG emissions. And the border carbon adjustment mechanism (CBAM) that is currently being put into place acts as a pricing of embodied GHG emissions at the border. For Switzerland, National Council member Gerhard Pfister proposed a policy package similar to what I consider here (<https://www.parlament.ch/de/ratsbetrieb/suche-curia-vista/geschaefte?AffairId=20220451>). It was not met with enthusiasm by the Council of States, but may resurface when the discussion turns to climate policy beyond the year 2030 (<https://www.tagesanzeiger.ch/klimaabgabe-abgelehnt-gerhard-pfisters-plan-geht-nicht-auf-543686918783>).

²¹As previously mentioned, this neglects medium to long term adjustments in both production and markets for labour and capital services. Similarly to the analysis of national CO₂ pricing, an analysis that allows for medium to long term adjustments to global GHG pricing would probably yield results that are skewed to slightly higher progressivity compared to what I find here, but similar heterogeneity within income groups (compare to findings by Landis, 2019; Rausch et al., 2011).

1611 than they are refunded and vice-versa for high-income households. This shows that –
1612 in analogy to national CO₂ pricing – global GHG pricing can be turned into an overall
1613 progressive policy if emissions pricing revenue is redistributed in an appropriate way.
1614

1615 While the upper panel shows averages only and absolute numbers in CHF, the lower
1616 panel of Figure 12 shows the distribution of net impacts (refunds minus direct costs)
1617 across households within decile groups and measures them relative to the baseline
1618 expenditure budget. This rendering shows that, in terms of averages (depicted by
1619 circles in the figure), the net benefits relative to household spending are larger for
1620 low-income households than the net costs for high-income households. It also shows
1621 that, due to large heterogeneity within income decile groups, the households with the
1622 highest relative net cost are still situated in the lowest income decile group.
1623

1624 The findings can be summarized similarly for pricing of global GHG emissions as
1625 they do for national CO₂ emissions. The visible direct cost of emissions pricing to
1626 households look regressive (this assessment is for the short term; in the medium to
1627 long term, indirect effects on markets for labour and capital services may neutralize
1628 the regressivity). Recycling of the revenues from emissions pricing can counteract the
1629 (perception of) regressivity in direct costs if properly designed. Per-capita lump-sum
1630 recycling makes overall policy distinctly progressive, but heterogeneity in low-income
1631 deciles means that hardship cases among the group of lowest-income households may
1632 still occur.
1633

1634 1635 1636 1637 1638 1639 1640 1641 1642 1643 1644 1645 **Additional considerations** 1646

1647 So far the analysis was based on the relation between policy cost and the base-
1648 line expenditure budget. Another issue often raised with less wealthy households and
1649 individuals is that they are restricted in their possibilities to invest due to credit con-
1650 straints. In a somewhat similar vein, households who rent their dwelling have little
1651 influence over the heating technology and insulation standards employed in it. These
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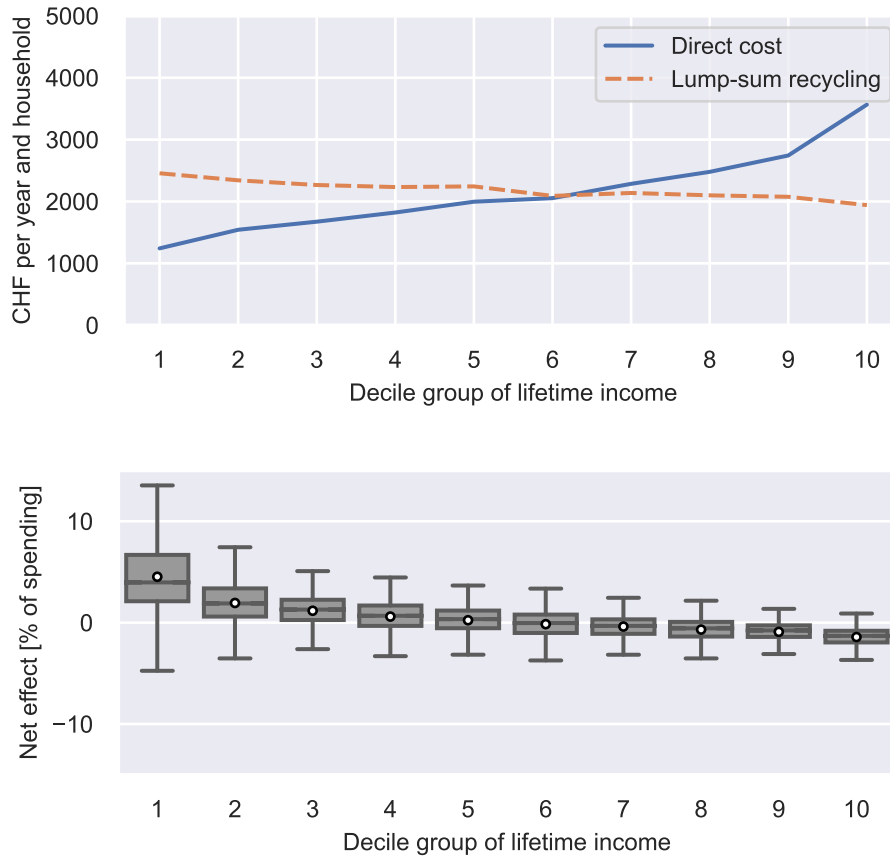


Fig. 12: Short term effects of global GHG pricing for Swiss consumers. Top: Direct cost of GHG pricing (blue solid) and per-capita lump-sum recycling (dashed orange). The lines represent averages across households within decile groups. Bottom: Net cost (direct cost minus lump-sum recycling) relative to baseline household expenditures. Box plots describe the distributions of net cost within the ten decile groups, the circles denote mean net cost within the groups.

two issues may hamper the ability of renters and lower-income house owners with little wealth (e.g. house owners with big mortgages) to efficiently react to the current CO₂ levy on heating fuels.

If we do not want to risk the CO₂-levy being restricted in its effectivity to (wealthy) house owners, we should make sure that everybody can react efficiently to the incentives set by the CO₂-levy. For credit constrained house owners, marketing schemes

1703 like 'Mietkauf' may be a valid solution.²² Under such schemes, house owners pay for
1704 the energy services over time and providers pay for the upfront investment for instal-
1705 lations of efficient technologies. Information about them should be promoted and a
1706 healthy competition between providers should be ensured. For renters, the situation
1707 is less straight forward. Market rents for housing should, in principle, account for the
1708 fact that efficient but more costly heating systems reduce heating costs and should
1709 be worth more rent. But not everybody is convinced that landlords are sufficiently
1710 aware of these trade-offs and that they make the efficient investments if they are not
1711 directly confronted with emissions pricing. The empirical evidence on this question
1712 seems mixed. Recent research suggest that if tenants are made aware of the energy
1713 cost savings, their willingness to pay for this through increased rents is commensu-
1714 rate (Lang & Lanz, 2021) and that market outcomes show little differences between
1715 flats that are offered for sale and those offered for rent in Germany (Singhal, Som-
1716 mer, Kaestner, & Pahle, 2023). Yet, in other contexts, financial incentives seem to
1717 have stronger effects for owner-occupied dwellings than for renter-occupied (Charlier,
1718 2015). One suggestion for addressing this potential problem is to either complement
1719 financial incentive with building standards and renovation requirements or to replace
1720 the incentives with such policies entirely (Charlier, 2015).

1733 In view of the fact that we cannot be certain that all actors in the economy can
1734 react efficiently to a CO₂-levy that focuses on heating fuels it may be regarded as
1735 suboptimal design that the CO₂-levy excludes motor fuels where such split-incentives
1736 problems do not exist. In addition to that, the distributional properties of pricing
1737 emissions from motor fuels are more favourable for low-income households than pricing
1738 emissions from heating fuels. To see this, compare bottom panels of Figures 9 and 8.
1739 A more in-depth critique of the exemption of transport from emissions pricing in
1740 Switzerland can be found in Landis, Rausch, and Kosch (2018).

1746
1747 ²²See, e.g., <https://www.energieheld.ch/renovation/finanzierung/mietkauf>

5 Concluding remarks

The analysis of how consumption based responsibility for GHG emissions, or GWP, is distributed across households allows for the following observations. First, where luxury goods are emissions-intensive, climate policy that focuses on these goods can ensure that high-income households make a relatively bigger effort to reduce emissions than low-income ones. Yet, if we were to restrict ourselves to regulating emissions related to luxury goods, the scope for significantly reducing emissions towards a net-zero target would be severely limited. Second, when emissions across the board are reduced, and if policy induced emissions reduction is costly, this will cost low-income households more (relative to expenditure budget) than high-income ones (regressive direct effects). Third, emissions pricing provides revenue for counteracting these regressive direct effects. The current design of the CO₂-levy recycles two thirds of the revenue to households in a per-capita lump-sum fashion. This makes the CO₂-levy overall progressive. Fourth, high heterogeneity in consumption patterns may result in hardship cases (even though their number can be kept in check by good policy design). Swiss society should be prepared to recognize hardship cases and support them. Fifth, the distributional properties of policies leading to costly emissions reduction are more favourable for emissions reduction in private transport compared to heating. The observation that there are split incentives between investors and users of clean technologies in the case of heating but not for private transport would make transport even more attractive for carbon pricing. The current CO₂-levy, however, only prices heating fuels and exempts motor fuels.

While the method used here for deriving GWPs from expenditure data and LCA is state-of-the-art, a few limitations have to be noted. Since the LCA that I used here does not differentiate consumption goods' quality levels by households of different income, there are possible differences in emissions per unit of consumption that may not be captured. In the case of public transport, I try to take this into consideration

1795 by denominating the LCA with person kilometres (pkm) and finding income dependent
1796 numbers of how far people ride per CHF spent on travel cards. For goods categories
1797 like restauration and clothing, the same was not done. Yet it is plausible that high-
1798 income household consume less per CHF spent in terms of physical units for both
1800 eating out and clothing than do low-income households. This may bias my analysis
1801 towards attributing too many emissions to high-income households and too few to low-
1802 income ones. Another drawback of my data sources is that the survey data contained
1803 in HABE are a few years old. No newer survey data are made available by the FSO,
1804 since the COVID-19 epidemic created irregularities in consumption patterns, and these
1805 are deemed to make the data unreliable. As a third caveat, it has to be noted that
1806 some observations about household expenditure patterns do not seem to represent
1807 “normal” monthly household expenditures. The HABE makes an effort to account for
1808 the fact that not all consumption goods are consumed every month of the year and
1809 asks households to give their *average* monthly expenditures based on annual bills for
1810 such consumption goods. Still some observed household patterns look “irregular”. This
1811 may be due to strong deviations from normal monthly values for consumption goods
1812 that are not reported as monthly averages based on annual bills or to the yearly values
1813 deviating from normal values. This means that some of the most extreme values in
1814 Figures 1–4 and 12 may represent atypical situations of the concerned households.
1815
1816 Finally, the observations made about the usefulness of revenue from emissions
1817 pricing for controlling the overall distributional impact of climate policy has to be
1818 qualified by the observation that such revenue may be abundant in the short term,
1819 when emissions are still high, but may dwindle even with rising emissions prices as
1820 emissions themselves go to zero in the longer term.
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Declarations	1841
	1842
Availability of data and materials. The GWP estimates used in this study are	1843
available in the Git repository “CCL-results”, https://github.com/OASES-project/	1844
CCL-results .	1845
	1846
	1847
The HABE data is available upon request from FSO and access to it is subject to	1848
signing a confidentiality agreement.	1849
	1850
	1851
The scripts and additional data for reproducing the findings and figures in this	1852
study are available in the Git repository “LCA_HABE”, https://github.com .	1853
	1854
	1855
Competing interests. The author declares that he has no competing interests.	1856
	1857
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	1861
Author contribution. FL analyzed, interpreted, and illustrated the data and wrote	1862
the manuscript.	1863
	1864
	1865
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sis, I relied heavily on the routines and methods published open source in Jakobs and	1868
Mutel (2023) by Arthur Jakobs and Chris Mutel . Arthur Jakobs kindly provided me	1869
with additional inputs to make the codes by Jakobs and Mutel (2023) fully operational.	1870
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Appendix A Additional detail for the data	1875
processing routine	1876
	1877
	1878
	1879
The combination of the LCA and household survey data in the HABE as described in	1880
Section 2 can mostly rely on harmonized units in the two data sets. That is, if GWP	1881
in the LCA is given in kg CO ₂ -eq per CHF spent or per kg or litre consumed, HABE	1882
provides consumption measured in the respective units. But some exceptions exist.	1883
	1884
	1885
	1886

These include heating, electricity, and public transport, where HABE gives expenditure in CHF but the LCA insists on physical units like Megajoule (MJ) or pkm travelled by mode of transport.²³ The following gives more detail on how these consumption categories are treated in my data processing routine.

Mobility: In the case of public transport modes (bus, tram, train), HABE gives households' expenditure for travel cards. Due to the fact that travel cards can allow for unlimited use of transport on certain lines or in certain areas over a given amount of time, it depends on the intensity of use of such travel cards how much pkm of transport services corresponds to the purchase price. This matters to the extent that different types of households may display different behaviours in terms of how intensively the travel cards are being used. Thus, the strategy is to find additional information about how far different household groups travel per CHF spent on different modes of transport. Within the groups, pkm travelled is taken to be proportional to spending.

The summary statistics of the Mobility and Transport Microcensus (Biedermann et al., 2017) provide information to this effect. Table "Tagesdistanz" in the "Verkehrshalten der Bevölkerung, Synthesetabellen"²⁴ contains daily distances for respondents in different household types. HABE can be used to obtain the expenditure on tickets and travel cards for different household types. The resulting ratios of pkm and expenditures then gives appropriate conversion ratios from CHF into pkm for different household types.

But which household types to differentiate? Table "Tagesdistanz" allows for taking subgroups of the Swiss population according to geographical region, language region, degree of urbanization, household composition, sex, age, household income, and status of employment. For combining with expenditure data from HABE, I consider the

²³My routine for combining LCA and expenditure data is driven to a large extent by the units used to denominate GWP in the LCA given by Jakobs and Mutel (2023). Other consumption categories such as restauration and fashion probably have varying GHG emissions per CHF spent for different household types. Yet, since the LCA gives GWP in kg CO₂-eq per CHF, I use this constant value across all differently household types.

²⁴Retrieved as file su-d-11.04.03-MZ-2015-T01.xls from <https://www.bfs.admin.ch/asset/de/2503927> on 27 February 2024.

dimensions geographical regions, language, household composition, and income. For deciding which dimension of household typification to focus on, I take averages of spending (in HABE) and the numbers for pkm (from the micro-census) and look at the correlation of the two along the dimensions. The dimension that gives the lowest correlation gives the most additional information complementary to estimating pkm as proportional to spending. I find the correlation to be the lowest along the dimension household composition and I conclude that household composition is the household property that provides the most additional information. I therefore consider household composition-specific ratios of pkm travelled per CHF spent as the conversion factors of expenditure to travel distance. Expenditure on multi-mode travel card expenditure is divided into distance travelled on different transport modes taking household composition-specific mode shares into account.

Heating and electricity: For housing related utility expenditures, households differ in their reporting detail. Some households can report all expenditures explicitly, others report expenditure for general utility expenditures that may include fees for waste disposal, waste water, fresh water, heating fuels, heat (from district or central heating), electricity, and “other”. My strategy is to split this general utility expenditure into the different subcategories according to the (average) expenditure shares of those households that report explicitly. For establishing the shares, I differentiate households that own their dwelling and those who rent.

For spending on waste disposal and energy, I make the following additional considerations. Waste is only taken out of general utility bills for those households that have no explicit expenditure for waste already. In the case of energy, I take heating and electricity out of the general utility bills if explicit energy expenditures is half the size of spending on general utility bills, taking this low ratio as an indicator, that explicit energy spending does not cover everything.

1979 For heating, even after splitting up general utility costs, approximately one third
1980 of households does not have any expenditures for heating. I fill in missing numbers in
1981 proportion to households' overall housing expenditures (rent or mortgage payments
1982 plus utilities) such that their average agrees to the averages of households that do
1983 report heating expenditures.
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1987 After these adjustments to spending records, I need to convert expenditures into
1988 physical quantities where the LCA units require it for assessment of embodied GHG
1989 emissions. This is the case for energy carriers like heating fuels, electricity and heat
1990 from district and central heating (in MJ), but also for waste (number of waste bags;
1991 cubic metres of waste water) and water (in cubic metres). For waste, waste water,
1992 and water, I use the price estimates used by Froemelt et al. (2018) that were kindly
1993 provided by Jakobs and Mutel. These price estimates are differentiated by region (eight
1994 cantons plus rest of Switzerland) and household size.
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2000 For energy, I take different reasons for differing prices into account. On the one
2001 hand, energy prices may vary over time, on the other hand, non-linear price plans
2002 make the average electricity price dependent on the volume consumed. For electric-
2003 ity, I use ElCom electricity tariffs per canton, year, and user profile (annual energy
2004 consumption).²⁵ For heating, rather than using time dependent price data, I use aggre-
2005 gate Swiss heating demand for the years 2015–2017 in MJ and distribute that across
2006 households were surveyed in the respective years in proportion to their expenditures
2007 for heat.
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2024 ²⁵ Available at, e.g., <https://www.visualize.admin.ch>

Appendix B Income and other household types

I find that lifetime income (viz. expenditures) is an important determinant of how much GWP different household types are responsible for. When I differentiate Swiss households along other socio-economic dimensions, I can create further groups of households with differing GWPs but this may be confounded by the fact that these new groups themselves have different lifetime income on average. In the following I try to make sure that the GWP patterns across different household types are not merely due to the different household types having different lifetime incomes, but that they reflect genuinely different consumption behaviour of the household types.

Figures B1–B3 show the separation of households into the two statuses of house ownership, into different household compositions, and into different degrees of urbanization – but this time within each decile group. Since the differentiation within decile groups yields the similar trends as does the differentiation within the whole population of surveyed households, I conclude that the differences in GWP across household types are not mainly driven by correlation between household types and lifetime income, but there is one exception to this.

Households living in a peri-urban setting, in the total of surveyed households, are responsible for the highest level of GWP (Figure 4). But within single decile groups (so somewhat controlled for income), we see that it's actually the households living in rural areas that have the highest level of GWP caused by their consumption (Figure B3).

The remaining trends observed across all surveyed households (Figure 4) remain true when controlling for income: Households living in urban areas are responsible for the least amount of GWP (Figure B3), households who rent are responsible for smaller amounts of GWP than those who own their residence (Figure B1), and larger households, on a per-capita basis, cause less GWP with their consumption than do smaller ones (Figure B2). This rather clear trend, observed when controlling for income, is attenuated a somewhat in Figure 4, however, indicating that bigger households tend

2071 to be of higher lifetime income which additionally increases their GWP when not
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2073 comparing for income.

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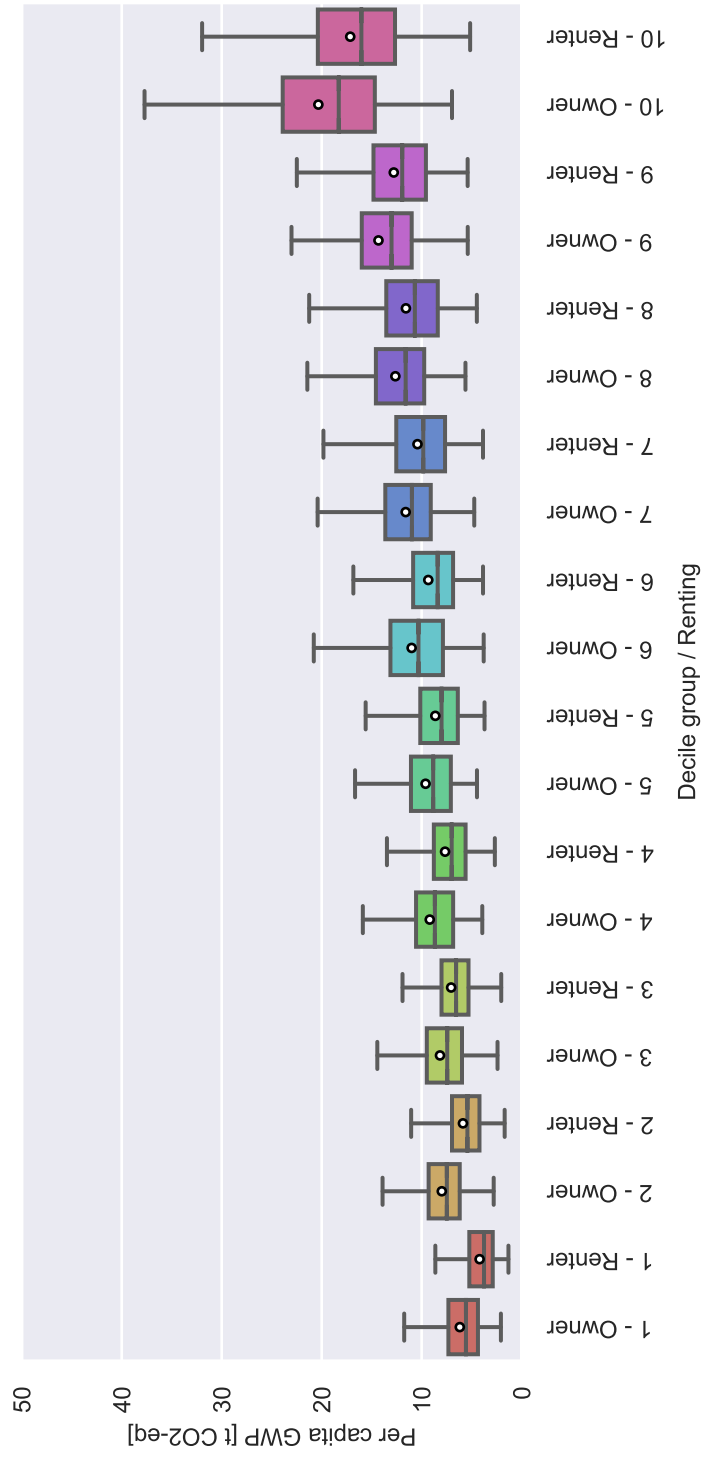


Fig. B1: GWP grouped by house ownership and by decile group of lifetime income.

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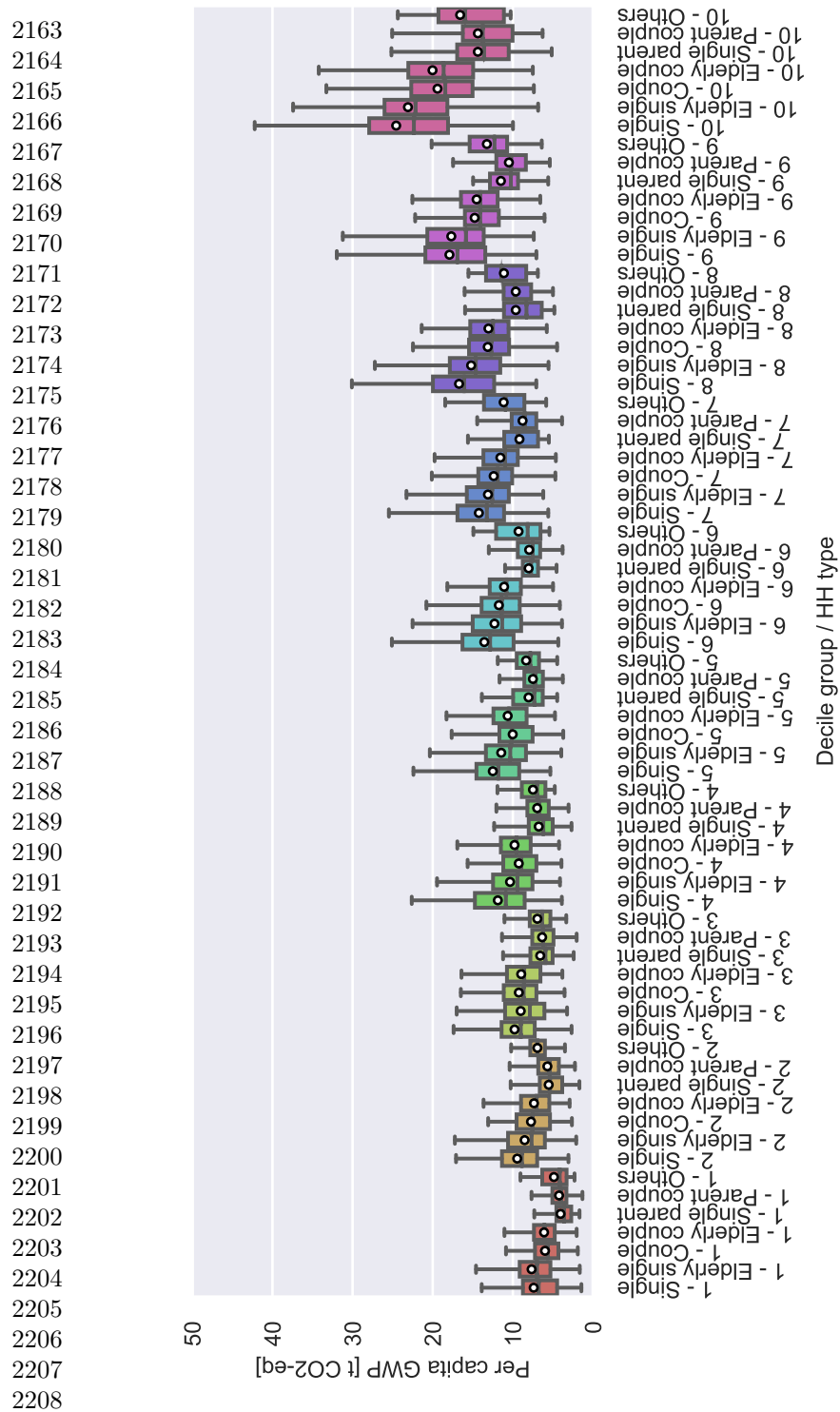


Fig. B2: GWP grouped by household composition and by decile group of lifetime income.

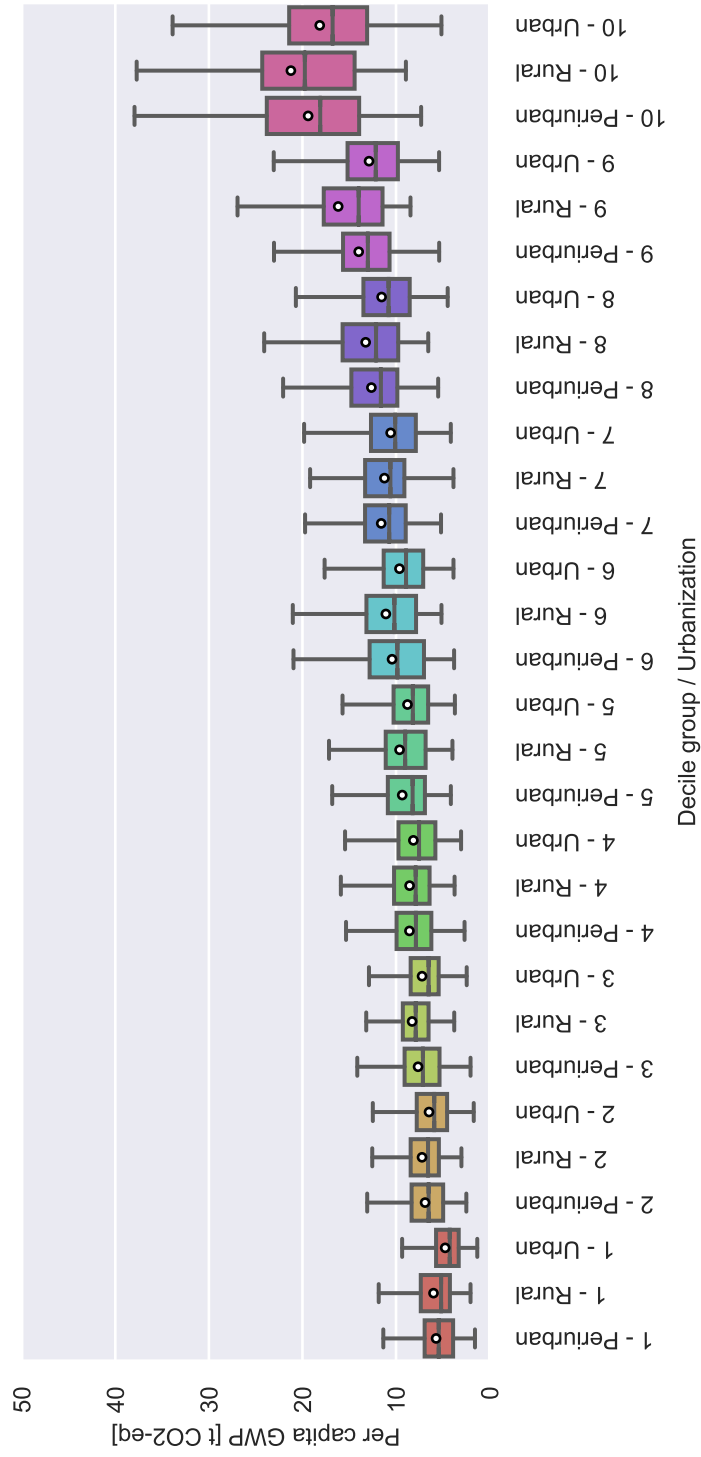


Fig. B3: GWP grouped by degree of urbanization and by decile group of lifetime income.

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Appendix C Confidence in found trends – comparing trends across decile groups and across 2-percentile-groups

Several trends for the means of per-capita GWP and GWP intensity have been observed in Section 3. The confidence with which I can say that these trends are real and not just due to a peculiar set of observations depends on the confidence intervals that I attach to these means and thus on the number of observations that are available for a given part of the population in HABE. With around 10'000 observations in the whole sample, I have around 1000 observations per decile group, whereas for 2-percentile-groups the number is approximately 200. I therefore have to expect tighter confidence intervals around decile group means and wider ones around 2-percentile-group means, giving us higher confidence in trends that we observe across decile groups than in trends across 2-percentile-groups.²⁶

This intuition about the size of confidence intervals is by and large met by the results in Figures C4 and C5, which show the confidence intervals for the means in Figures 2 and 3, respectively. Figure C4 shows the situation for GWP intensity in kg CO₂-eq per CHF. Here, the trend across decile groups is clear in the sense that we observe a strictly monotone decrease in lifetime income. The confidence in the trend is not perfect but still distinct: The confidence intervals for means in neighbouring decile groups overlap, but at least confidence intervals between one decile group and its second-nearest neighbours are disjoint. For 2-percentile groups, no monotone trend can be affirmed, most confidence intervals overlap with several others, but we can say with confidence, that the mean GWP intensities of the two lowest income 2-percentile-group are higher than the those of the other 2-percentile-groups.

²⁶Confidence intervals around the mean m are determined as $m \pm 1.96 \cdot s / \sqrt{n - 1}$, where s is the (weighted) sample standard deviation and n the number of observations in said sample.

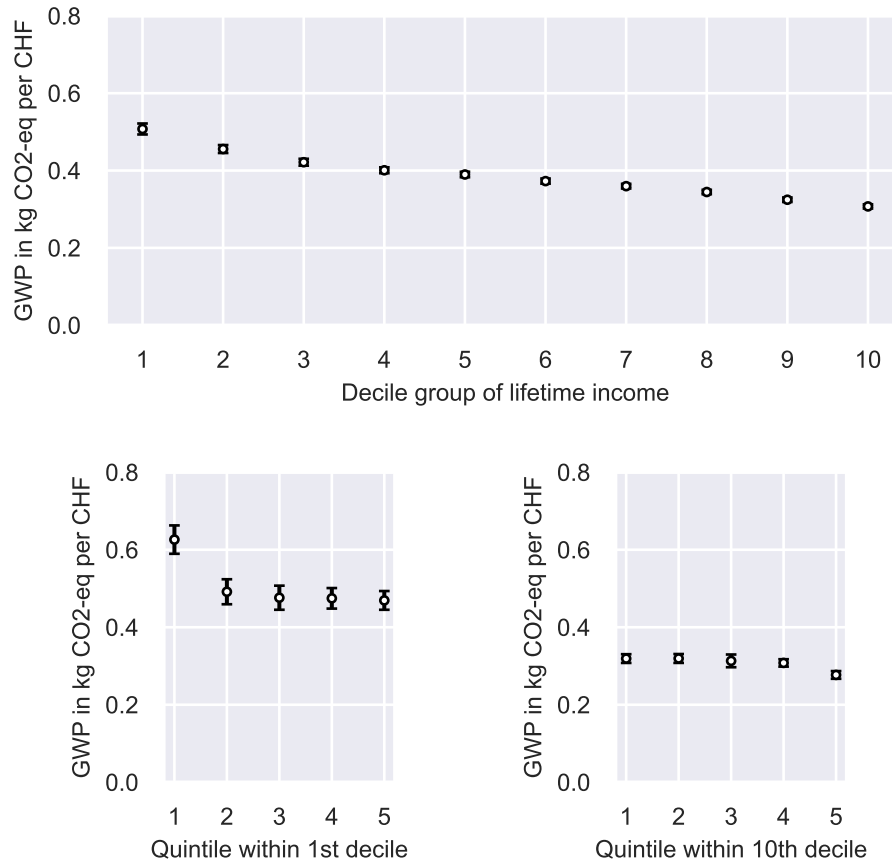
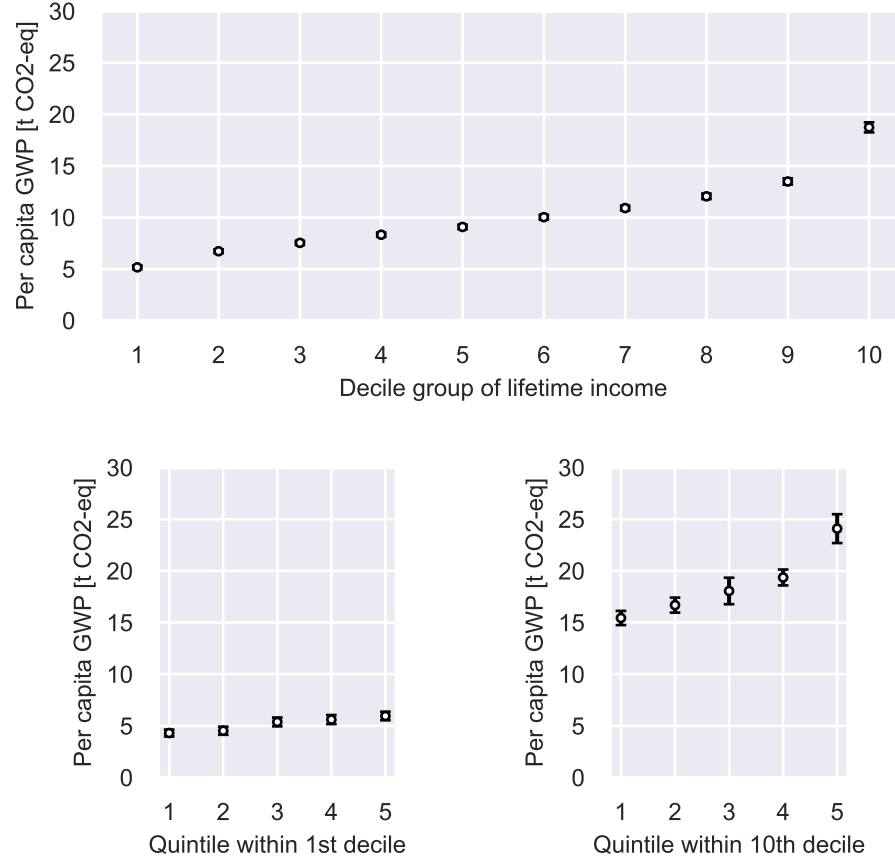


Fig. C4: 95 percent confidence intervals for decile group means of GWP intensity of consumption (top panel) and for 2-percentile-group means (bottom panels).

The trend for per-capita GWP across decile groups in the top panel of Figure C5 is also clear: Mean per-capita GWP increases with income and we are highly confident in our observation since the confidence intervals of the ten means never overlap. For the 2-percentile-group numbers in the bottom-panel the situation is less clear. Even though we can suspect a monotone trend, the confidence intervals of numerous pairs of means overlap and I cannot be confident of the trend. What I can say with some confidence, however, is that the last 2-percentile-group (the one with the highest lifetime income) has the highest per-capita GWP.

2347 These considerations are based on variation in consumption patterns alone. They
 2348 do not include uncertainties about the GWP associated with consumption categories
 2349 or about variation of this per-unit GWP across different household types.



2381 **Fig. C5:** 95 percent confidence intervals for decile group means of per-capita GWP
 2382 (top panel) and for 2-percentile-group means (bottom panels).

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